■ OPERATING SYSTEMS – EXAM CHECKLIST

■ SU1 – Introduction to Operating Systems

- ■ Understand Boot Procedure (with file system diagram and step numbers).
- ■ Know comparison of OS types (MCQ only).
- ■ Basic idea of System Calls (MCQ only).
- ■ Know different Kernel types (no direct question).
- ■ Ignore OS history and background theory.

■ SU2 – Processes & Threads

- ■ Understand Process Model and 3-State Diagram (Ready, Running, Blocked).
- ■ Identify valid/invalid states/arrows in variations.
- ■ Explain numbered process transitions (as in mid-sem).
- ■ Apply process model to real-world examples.
- ■ Understand types of threads (user vs kernel).
- ■ Know multithreading diagrams conceptually.

■ SU3 – Interprocess Communication (IPC)

- ■ Expect a Race Conditions question.
- ■ Study all synchronization techniques (Strict alternation, Interrupts, Locks, Peterson, Semaphores, Mutexes).
- ■ Be able to name and describe each technique (no marks for name only).

■ SU3 – CPU Scheduling & PRA

- ■ Know all algorithms: FCFS, SJF, SRTN, Priority, Round Robin, Second-Chance (Clock).
- ■ Practice CPU scheduling cycles carefully – marking is strict.
- ■ PRA: Focus on Second-Chance Algorithm (show only hits & faults).
- ■ One mistake per row = zero marks – be accurate.

■ SU4 – File Management & Boot Process

- ■ Study file management diagram and explain it.
- ■ Explain what happens when a block fails.
- ■ Know boot procedure + file system combination (one of these will appear).

■ SU5 – Input/Output Management

- ■ Understand DMA deeply (adapt to NVMe + GPU scenarios).
- ■ Know DMA diagram connections and flow.
- ■ Understand Programmed I/O vs Interrupt I/O differences (MCQ or table questions).

■ SU6 – Deadlocks

- ■ Know what a Deadlock is and how to detect, prevent, and avoid it.
- ■ Understand difference between Prevention and Avoidance.
- ■ Study Kaufman conditions and their solutions.
- ■ Be able to identify and explain deadlock in diagram form.

■ SU7 – Security

- ■ Understand how attacks like Buffer Overflow or SQL Injection work.
- ■ Know 3 common security mistakes and how to prevent them (e.g., single-factor → MFA).
- ■ Ignore Copyright section.

■ Not Examined Topics

- ■ OS History and background evolution.
- ■ Detailed memory without abstraction (base/limit registers).
- ■ RAID.
- ■ Multiprocessor memory management diagrams.
- ■ Bitmap/linked list allocation methods.
- ■ Long block diagrams with 'X's.
- ■ Memory management past PRA.
- ■ Copyright.